

Sprint 2 Intermediate Product Report

Team Name: AI-Driven Robotic Manipulation: Integrating Learning Models with Real-World Control

Team Members: Toan Pham, Albert Vo

Sponsor: Dr. Nakul Gopalan

Date: Oct 7, 2025

1. Overview

Sprint Goal: Integrate panoptic scene graph generation and demonstrate improved policy performance

Work Completed:

SBI-201 – VLPrompt Implementation (In Progress)

2. Achievements

User Stories:

US-004: VLPrompt Implementation – In Progress

3. Risk Management

New Risks Identified

Risk D: Scene Graph Quality

Status: Active

Probability: Medium

Impact: High - Core to approach

Mitigation: Manual annotation backup (20 hours allocated)

Owner: Albert Vo

Risk E: VLPrompt Latency

Status: Monitoring

Probability: Medium

Impact: Medium - May limit real-time use

Mitigation: Caching, batch processing, model distillation

Owner: Toan Pham

Resolved Risks

Risk A (MuJoCo): Resolved with institutional license

Risk C (Data Quality): Improved with better interface

4. Contribution Metrics

Velocity: 28 story points (of 35 planned)

Task Completion: 21 of 28 tasks (75%)

GPU Hours Used: 96 hours

Demonstrations Collected: 6 (target: 8)

Scene Graphs Generated: 8,500 frames

Model Performance: 48% success rate (↑13% from baseline)

5. Evidence

Taiga Board: See project board for sprint tasks and burndown.

GitHub Repository: See commits and PR history for sprint work.

6. Feedback Summary

Progress Check

VLPrompt setup complete, debugging generation

SAM integration functional, optimizing speed

6/8 demonstrations collected

Scene graph conditioning architecture designed

Adjustments Made

Reduced target to 8 demonstrations

Added caching for scene graph generation

Increased Sol GPU allocation request

Sponsor Feedback

"Good progress on scene graphs. Focus on quality over quantity for data. The VLPrompt choice is sound - it aligns with my recent work on language grounding. Consider adding failure case analysis."

7. Updates

Added caching for scene graph generation

Increased Sol GPU allocation request

Sponsor Feedback

"Good progress on scene graphs. Focus on quality over quantity for data. The VLPrompt choice is sound - it aligns with my recent work on language grounding. Consider adding failure case analysis."

8. Signature from the Product Owner/Client

Signature/email:

A handwritten signature in red ink, consisting of a large, stylized 'X' or 'A' shape followed by a horizontal line.